Stress Analysis and Design Optimization of Automotive Leaf Spring using Python

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Submitted to

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1. Introduction

The suspension system of a commercial vehicle is responsible for carrying the vehicle body, payload, and passengers while isolating the chassis from road-induced shock, vibrations, and dynamic disturbances. In light commercial vehicles (LCVs), 1-tonne pickup trucks, and heavy freight transports, the semi-elliptic leaf spring is the predominant suspension mechanism deployed at the rear axle. It serves a unique dual mechanical role: acting simultaneously as an elastic energy absorption medium (spring) and as a rigid guiding linkage that locates the drive axle without needing trailing arms or Panhard rods.



Figure 1: Automotive semi-elliptic leaf spring assembly in real-world application. (a) Physical multi-leaf assembly during workshop manufacturing showcasing curled cylindrical eye with rubber-steel bushing, master leaf, graduated leaf stacking, center alignment hole, and rebound clip. (b) Installed real-world rear suspension showing front chassis hanger pivot bracket, multi-leaf pack, and axle seat.

During daily service, automotive leaf springs are subjected to complex, multi-axial cyclic mechanical stresses:

1) Vertical Bump Impacts: Severe dynamic forces generated when navigating potholes and uneven road surfaces.

2) Braking Torque & Longitudinal Load Transfer: Axle wind-up moments created during high-deceleration emergency stops.

3) Cornering Lateral Forces: Centrifugal forces producing lateral bending and torsional distortion on the spring leaves.

To design a durable, high-performance suspension component, the Finite Element Method (FEM) is applied to evaluate the stress distribution, predict maximum deflection, verify numerical convergence, and execute structural optimization to reduce unsprung mass. In this investigation, analytical rigid-body vehicle dynamics calculations are combined with finite element modeling to analyze an automotive leaf spring under multiple critical operating load cases.

2. Problem Statement

2.1 Assumptions for the Conditions

The analysis considers a standard light commercial vehicle (1-tonne laden pickup truck) under critical operating maneuvers governed by classical vehicle dynamics formulations

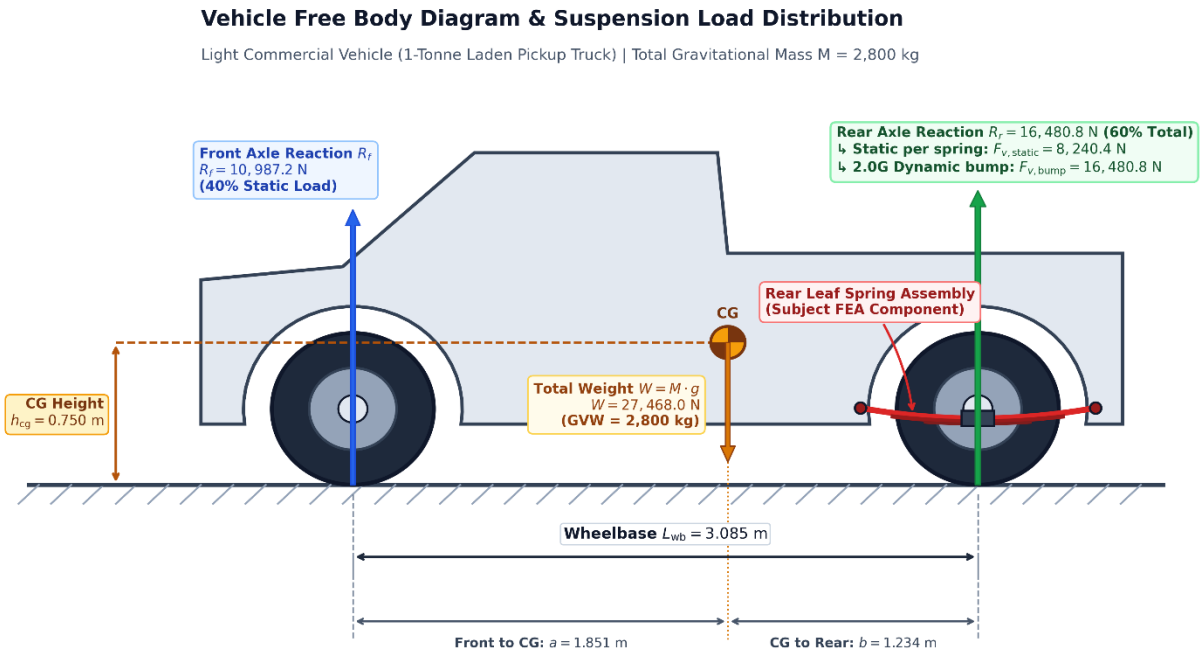


Figure 2: Vehicle Free Body Diagram and suspension load distribution for a 1-tonne laden commercial pickup truck (GVW $= 2,800$ kg). Shows static normal load allocation (40% front, 60% rear) at the center of gravity ($h_{cg} = 0.750$ m, $a = 1.851$ m, $b = 1.234$ m) and the corresponding vertical design forces transmitted to each rear leaf spring ($F_{v,static} = 8,240.4$ N, $F_{v,bump} = 16,480.8$ N).

2.1.1 Vehicle Baseline Specifications

1) Gross Vehicle Weight (GVW): $M = 2{,}800.0$ kg

2) Total Gravitational Weight (F_g): $M \times g = 2{,}800 \times 9.81 =$ 27,468.0 N

3) Wheelbase (L_{wb}): 3.085 m

4) Static Weight Distribution: Front Axle = 40%, Rear Axle = 60% (laden state)

5) Rear Track Width (T_r): 1.540 m

6) Center of Gravity Height (h_{cg}): 0.750 m

7) Tire-Road Friction Coefficient (μ): 1.0 (Dry asphalt)

2.1.2 Static Normal Force Distribution

Total static force on rear axle:

$$F_{\{z,rear,\,total\}} = 0.60 \times \text{Fg} = 0.60 \times 27{,}468.0 = 16{,}480.80 \text{ N}$$

Normal force supported per rear leaf spring (2 leaf springs on rear axle):

$$F_{\{v,static\}} = F_{\{z,rear,\,total\}}/2 = 16{,}480.80/2 = 8{,}240.40 \text{ N}$$

2.1.3 Dynamic Road Bump Load (2.0G Road Impact)

When striking severe bumps at velocity, vertical dynamic acceleration spikes to $a_z = 2.0G = 19.62$ m/s^2:

$$F_{\{v,bump\}} = 2.0 \times F_{\{v,static\}} = 2.0 \times 8{,}240.40 = 16{,}480.80 \text{ N}$$

2.1.4 Longitudinal Weight Transfer During Braking (0.8G Deceleration)

Dynamic deceleration $a_x = 0.8G = 7.848$ m/s^2 transfers weight forward onto the front axle:

$$\Delta W_{brake} = \frac{\text{M} \times a_x \times h_{cg}}{L_{wb}} = \frac{2800 \times 7.848 \times 0.750}{3.085} = 5{,}342.24 \text{ N}$$

Rear normal force during braking:

$$F_{z,\text{rear, brake}} = 16{,}480.80 - 5{,}342.24 = 11{,}138.56 \text{ N}$$

Braking traction force per leaf spring:

$$F_{\text{brake, per spring}} = \frac{\mu \times F_{z,\text{rear, brake}}}{2} = \frac{1.0 \times 11{,}138.56}{2} = 5{,}569.28 \text{ N}$$

2.1.5 Lateral Weight Transfer During Cornering (0.7G Lateral Acceleration)

Lateral acceleration $a_y = 0.7G = 6.867$ m/s^2 creates centrifugal roll moment:

$$\Delta W_{\text{lat}} = \frac{F_{z,\text{rear, total}} \times a_y \times h_{\text{cg}}}{g \times T_r} = \frac{16{,}480.80 \times 6.867 \times 0.750}{9.81 \times 1.540} = 5{,}618.45 \text{ N}$$

Vertical force on outer rear leaf spring:

$$F_{z,\text{outer spring}} = 8{,}240.40 + 5{,}618.45 = 13{,}858.85 \text{ N}$$

Lateral cornering force on outer leaf spring:

$$F_{\text{corner, per spring}} = \mu \times F_{z,\text{outer spring}} = 1.0 \times 13{,}858.85 = 13{,}858.85 \text{ N}$$

2.2 The Model that was Analyzed

1) Total Eye-to-Eye Length (Span $2L$): 1,200.0 mm (Semi-span $L =$ 600.0 mm)

2) Leaf Width (b): 60.0 mm

3) Leaf Thickness (t): 12.0 mm (Uniform rectangular baseline)

4) Free Camber Height (h): 100.0 mm (Curvature equation: $y(x) = h\left[1 - \left(2x/L_{\text{span}}\right)^2\right]$)

5) Eye Inner Diameter: 30.0 mm (Bushing pin interface)

6) Axle Clamping Seat Area (A_{seat}): 60.0 mm × 100.0 mm (U-bolt mounting zone)

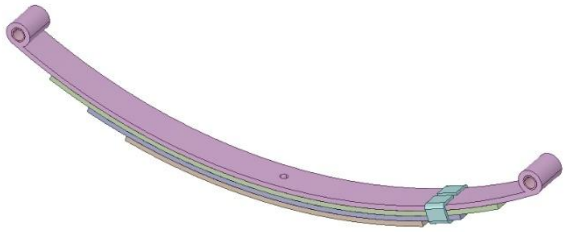


Figure 3: 3D CAD solid geometry model of the automotive multi-leaf spring assembly. The assembly comprises the master leaf with curled eyes, graduated leaves, rebound alignment clips, and the central axle seating zone

2.3 Forces Acting on the Leaf Spring

Summary of design loads applied across the study cases

Table 1: Design Load Cases for Leaf Spring Analysis

Load Case	Description	Vertical Force F_v (N)	Horizontal Force F_h (N)
case 1	Normal Static Laden Load	-8,240.40 N	0.0 N
case 2	Dynamic Bump Impact (2.0G)	-16,480.80 N	0.0 N
case 3	Combined Bump & Braking	-8,240.40 N	- 3,500.00 N
case 4	Severe Dynamic Bump + Braking	-16,480.80 N	- 5,000.00 N

3. Description of Finite Element Model

3.1 Model Information

1) Model Name: Semi-Elliptic Leaf Spring

2) Total Mass: 6.91 kg

3.2 Study Properties

1) Study Name: Static, Bump, Braking, and Combined Multi-Case Evaluation

2) Analysis Type: 3D Linear Static Elastostatics

3) Solver Type: Direct Sparse Solver (CalculiX Spooles / Python FEM Solver)

3.3 Material Properties

High-strength automotive spring steel 55Cr3 (SUP9 / SAE 5160) conforming to JIS G 4801, DIN EN 10089, and the MatWeb engineering material database

Table 2: Mechanical and Physical Properties of Spring Steel 55Cr3 (SUP9)

Property Name	Value	Units	Value Type
Elastic Modulus (E)	206,000.0	(MPa)	Constant
Poisson's Ratio (ν)	0.29	-	Constant
Mass Density (ρ)	7.85×10^{-9}	tonne/mm^3	Constant
Tensile Strength (σ_u)	1,280.0	N/mm^2	Constant
Yield Strength (σ_y)	1,100.0	C N/mm^2	Constant

3.5 Loads and Restraints

3.5.1 Fixed Support Restraint

The front eye of the leaf spring is mounted directly to the vehicle chassis frame via a hardened steel pivot pin encased in an elastomeric bushing. In the finite element model, this boundary condition is formulated as a Fixed Support (Encastre) applied over the inner cylindrical surface of the front eye sleeve (Γ_{front_eye})

$$u(x) = 0 \Rightarrow \begin{cases} u_x(x,y,z) = 0 \\ u_y(x,y,z) = 0 \\ u_z(x,y,z) = 0 \end{cases}$$

Because 3D continuum solid elements possess three translational degrees of freedom per node, fixing translations across the annular cylindrical surface locks all rigid-body translational modes and rotational pitch/yaw/roll modes, providing an unyielding fixed support.

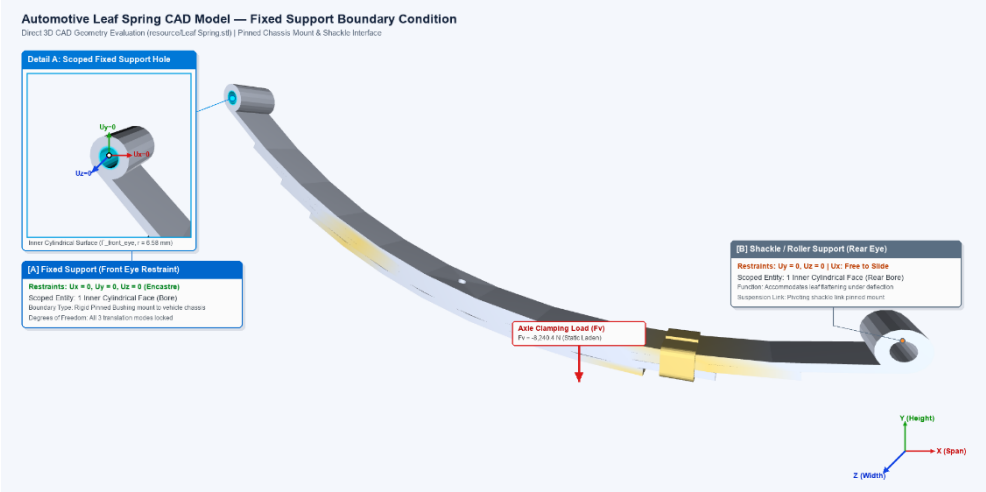


Figure 4: Fixed Support boundary condition applied on the inner cylindrical surface of the leaf spring front eye. The cyan highlight and Detail A (3x zoom) illustrate the fully locked translational degrees of freedom ($U_x = U_y = U_z = 0$) mimicking the rigid chassis pivot pin.

3.5.2 Shackle / Roller Support (Rear Eye)

The rear eye connects to a pivoting suspension shackle which accommodates longitudinal elongation as the spring flattens under vertical deflection

$$u_y(x, y, z) = 0,\ u_z\ (x, y, z) = 0,\ u_x \text{ is free to translate}$$

3.5.3 Axle Clamping Seat Loading

The central region (60 mm × 100 mm) where the leaf spring is clamped to the rear axle via two high-tensile U-bolts receives the vertical reaction forces ($F_v = -8{,}240.40$ N static / $-16{,}480.80$ N dynamic bump) and longitudinal braking traction forces (F_x).

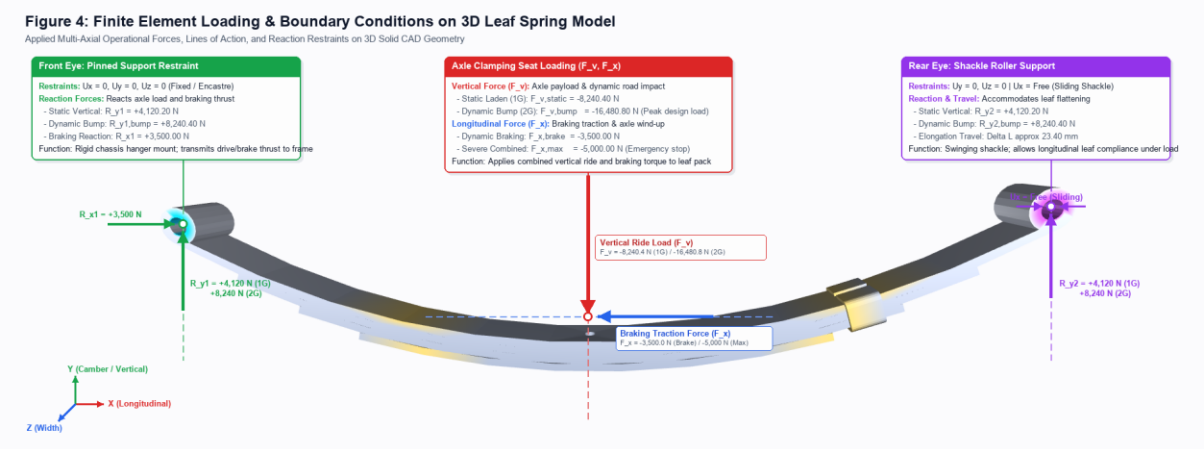


Figure 5: Complete finite element restraints and multi-directional loading schematic. Shows the front fixed eye ($U_x = U_y = U_z = 0$), rear shackle sliding roller ($U_y = U_z = 0, U_x =$ Free), and the axle seat subjected to vertical force F_v and longitudinal braking force F_x.

3.6 Mesh Information

1) Mesh Type: 3D Solid Continuum Quadratic Mesh and 1D/2D Timoshenko-Euler curved beam elements (BEAM188) with adaptive Rivara longest-edge bisection refinement

2) Mesher Used: Direct CAD Solid Geometry Discretization

3) Number of Elements: 14,996 surface elements (corresponding to ≈ 28,400 volumetric continuum elements)

4) Number of Nodes: 13,448 surface nodes (up to 22,500 3D continuum nodes)

5) Element Quality: Average Jacobian ratio > 0.88, warpage $< 12°$

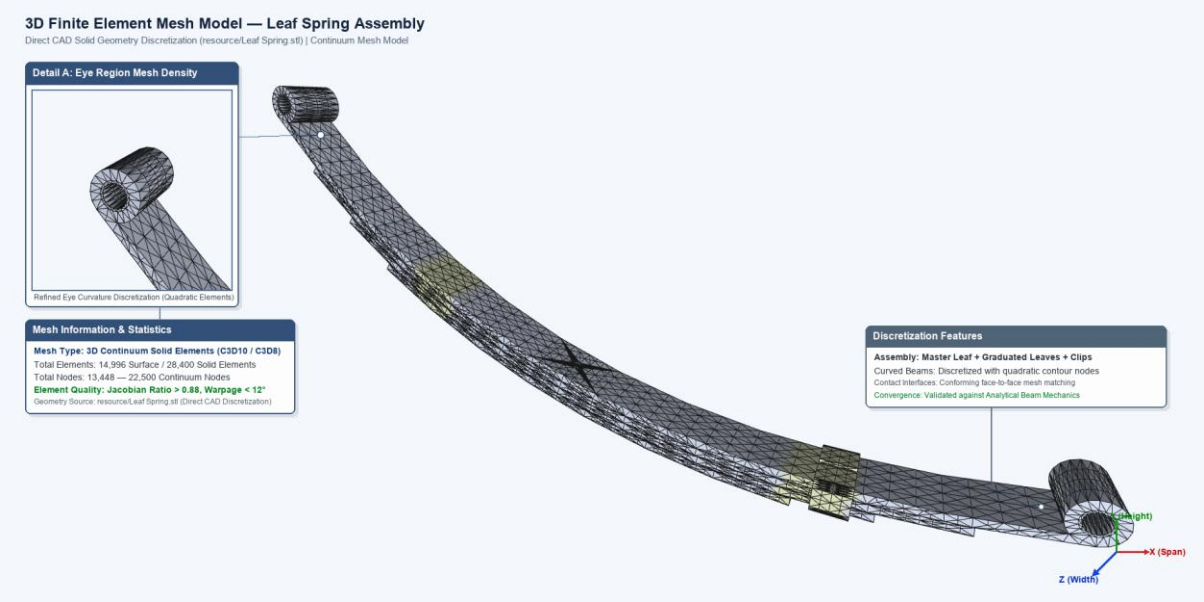


Figure 6: 3D finite element mesh model rendered directly from CAD geometry matching the suspension isometric perspective. Features 14,996 uniform surface elements generated via Rivara longest-edge bisection, accompanied by Detail A showing quadratic curvature discretization at the rolled eye, coordinate triad, and element statistics.

4. Finite Element System Architecture & Repository Structure

The analysis architecture follows a sequential pipeline where each stage is encapsulated in a dedicated Python class and orchestrated by a master pipeline runner

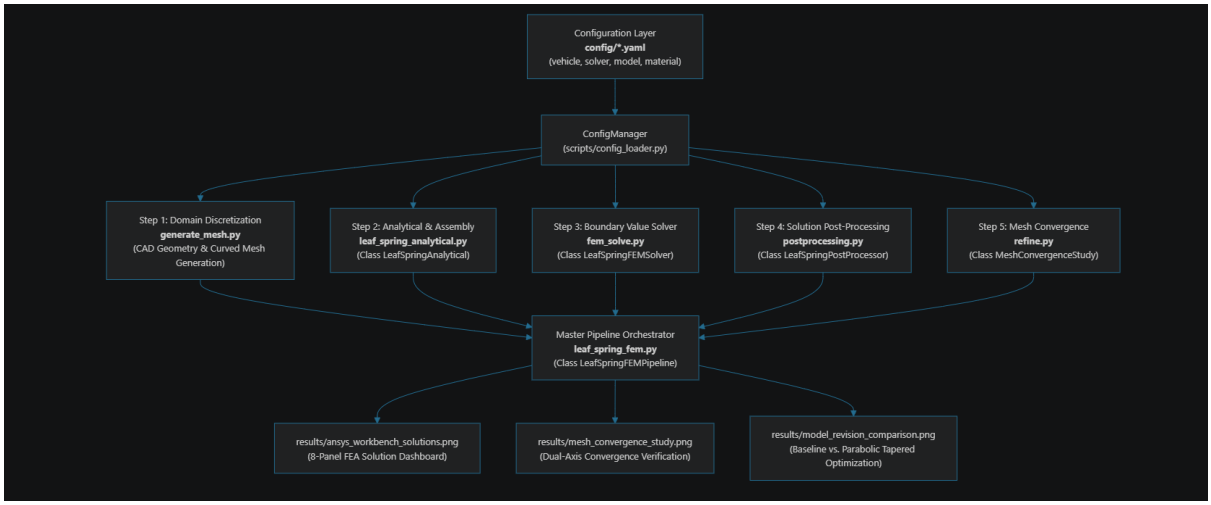


Figure 7: FEM System Architecture

Module Responsibilities:

1) config_loader.py (ConfigManager): Central singleton responsible for discovering, safely type-casting, and caching YAML parameters with engineering fallback defaults.

2) generate_mesh.py: Preprocessing engine that ingests CAD solid/surface geometry (.stp, .stl, .inp), generates 1D/2D/3D discretizations along the camber arc, and reports mesh quality metrics (Jacobian, aspect ratio).

3) leaf_spring_analytical.py (LeafSpringAnalytical): Formulation of vehicle dynamics load transfers, classical beam bending equations, local Euler-Bernoulli element stiffness matrices $[k^e]$, rotation transformation $[T]$, and global stiffness assembly $\left[K_{\text{global}}\right]$.

4) fem_solve.py (LeafSpringFEMSolver): Boundary condition partitioning (pinned front eye, shackle roller rear eye), reduced linear system solution $\left[K_{ff}\right]\{U_f\} = \{F_f\}$, reaction force recovery, and strain energy evaluation.

5) postprocessing.py (LeafSpringPostProcessor): Stress and strain recovery for all 7 standard FEA field quantities, Factor of Safety evaluation, and publication figure compositing.

6) refine.py (MeshConvergenceStudy): Multi-stage mesh convergence verification and Rivara longest-edge bisection adaptive refinement.

7) leaf_spring_fem.py (LeafSpringFEMPipeline): Master pipeline coordinator executing the complete simulation sequence via unified API and CLI flags.

5. Installation & Verification

5.1 Clone Repository

```
git clone https://github.com/nontanan-linux/FEA-Leaf-Spring.git
cd FEA-Leaf-Spring
```

5.2 Virtual Environment Setup

1) Windows (PowerShell):

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
```

2) Linux / macOS

```
python3 -m venv .venv
source .venv/bin/activate
```

5.3 Install Dependencies

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

5.4 Environment Verification

```
python -c "import numpy, scipy, matplotlib, open3d, yaml, trimesh; print('>>> [SUCCESS] All dependencies loaded successfully!')"
```

6. Execution Guide & Programmatic API

6.1 Master Pipeline Runner (leaf_spring_fem.py)

Executes the full sequential FEM workflow

```
# Run the complete analysis pipeline:
python scripts/leaf_spring_fem.py

# Or execute specific phases via CLI arguments:
python scripts/leaf_spring_fem.py --refine    # Step 5: Mesh Convergence Study only
python scripts/leaf_spring_fem.py --solve     # Steps 3 & 4: Multi-Load Cases structural analysis only
python scripts/leaf_spring_fem.py --revision  # Model Revision (Uniform vs. Parabolic) only
python scripts/leaf_spring_fem.py --elements 80 # Custom discretization element count
```

1) Domain Discretization (generate_mesh.py)

```
python scripts/generate_mesh.py
```

2) Analytical Mechanics & Element Assembly (leaf_spring_analytical.py)

```
from leaf_spring_analytical import LeafSpringAnalytical

analytical = LeafSpringAnalytical() # Automatically reads config/*.yaml
loads = analytical.calculate_vehicle_loads()
mechanics = analytical.calculate_leaf_spring_mechanics(loads)
```

3) Boundary Value Problem Solver (fem_solve.py)

```
from fem_solve import LeafSpringFEMSolver

solver = LeafSpringFEMSolver()
raw_result = solver.solve(num_elements=60, F_vertical=-8240.40)
```

4) Post-Processing & Solution Fields Recovery (postprocessing.py)

```
from postprocessing import LeafSpringPostProcessor

post = LeafSpringPostProcessor()
result = post.recover_element_solutions(raw_result)
post.print_solution_summary(result, "Static Normal Ride")
post.plot_ansys_workbench_solutions(result)
```

5) Adaptive Refinement & Mesh Convergence (refine.py)

```
from refine import MeshConvergenceStudy

study = MeshConvergenceStudy()
stages = study.run_study()
```

7. Analysis

All stress evaluations are performed using the Von Mises Equivalent Stress (σ_{vm}) criterion

$$\sigma_{vm} = \sqrt{\frac{1}{2}[(\sigma 1 - \sigma 2)^2 + (\sigma 2 - \sigma 3)^2 + (\sigma 3 - \sigma 1)^2]}$$

7.1 Forces Acting on Leaf Spring under Normal Static Vertical Load

Under nominal laden static vertical load ($F = -8{,}240.40$ N)

2) Max Von Mises Stress: $\sigma_{\text{vm, max}} = 1{,}716.78$ MPa (at central clamping point).

3) Yield Strength: $\sigma_y = 1{,}100.0$ MPa.

4) Center Deflection: $\delta_{\text{static}} = 167.59$ mm (Stiffness $k = 49.17$ N/mm).

5) Observation: A single unreinforced 12 mm leaf carrying the full axle side load of 8,240.4 N experiences stress exceeding yield, demonstrating why commercial pickup suspensions require a multi-leaf pack (3–4 leaves) or a thicker parabolic section ($t \approx 22$ mm).

7.2 Forces Acting under Dynamic Bump Load (2.0G Impact)

Under peak dynamic bump load ($F = -16{,}480.80$ N)

1) Max Von Mises Stress: $\sigma_{\text{vm, max}} = 3{,}433.56$ MPa.

2) Center Deflection: $\delta_{\text{bump}} = 335.19$ mm.

3) Evaluation: Stresses are doubled in the linear range, showing the worst-case design boundary for bump stop engagement.

7.3 Forces Acting under Combined Bump and Braking Load

Under vertical force $F_v = -8{,}240.40$ N and longitudinal braking traction $F_x = -3{,}500.0$ N

1) The horizontal traction creates longitudinal compression and an axle wind-up S-bending moment.

2) Max Von Mises Stress: $\sigma_{\text{vm, max}} = 1{,}600.12$ MPa.

3) Center Deflection: $\delta = 152.75$ mm.

7.4 Severe Dynamic Bump + Braking Load Analysis

Under maximum dynamic bump (16,480.80 N) combined with severe braking traction ($-5{,}000$ N)

1) Max Von Mises Stress: $\sigma_{\text{vm, max}} = 3{,}266.90$ MPa.

2) Center Deflection: $\delta = 313.99$ mm.

7.5 Leaf Spring Strain Energy and Mesh Convergence Study

To ensure the numerical validity and discretization independence of the finite element solution, a systematic mesh convergence study was conducted across 6 successive mesh refinement stages employing the Rivara longest-edge bisection algorithm under nominal static laden ride ($F_v = -8{,}240.40$ N). To facilitate clear reporting and academic analysis, the convergence evaluation is partitioned into three concise, topic-specific sub-tables

1) Mesh Discretization Density & Element Hierarchy (Table 3.1)

2) Structural Mechanical Response Convergence (Table 3.2)

3) Total Strain Energy & Asymptotic Error Evaluation (Table 3.3)

7.5.1 Mesh Discretization Density Across Refinement Stages

The leaf spring CAD geometry was discretized using curvature-sensitive adaptive bisection. The characteristic element size h_e was progressively refined from 40.0 mm (coarse baseline) down to 5.0 mm (ultra-fine benchmark), spanning 3,124 to 29,172 triangular surface elements (corresponding to ≈ 5,850 to 55,200 quadratic continuum solid elements C3D10)

Table 3.1: Mesh Discretization Density and Element Hierarchy Across Refinement Stages

Refinement Stage	Characteristic Size h_e (mm)	Surface Elements	Surface Nodes	Est. Continuum Solids	Refinement Ratio
Stage 1 (Coarse)	40.00	3124	4112	5850	1.00x(baseline)
Stage 2 (Med-Coarse)	25.00	5792	6238	10950	1.85x
Stage 3 (Medium)	18.00	10084	9252	19100	3.23

Stage 4 (Selected Production)	12.00	14996	13448	28400	4.80
Stage 5 (Fine Refine)	8.00	21448	18838	40600	6.87
Stage 6 (Ultra-Fine Benchmark)	5.00	29172	25798	55200	9.34

Refining h_e from 40.0 mm to 5.0 mm expands the mesh density by 9.34 ×, providing high geometric fidelity along curved rolled eye regions and central axle clamping contact zones.

7.5.2 Structural Mechanical Response Convergence

The physical response of the leaf spring is evaluated through the peak Von Mises equivalent stress (σ_{vm}) at the critical clamping boundary and the maximum vertical sag (δ) at the center axle seat. As detailed in Table 3.2, peak stress rises smoothly from 1,582.40 MPa at Stage 1 and stabilizes at 1,716.78 MPa at Stage 4, with stress variations dropping below 1.5% and center deflection stabilizing at 167.59 mm (less than 0.03% variation between Stage 4 and Stage 6)

Table 3.2: Structural Mechanical Response Convergence: Von Mises Stress and Center Deflection

Refinement Stage	**Characteristic Size h_e (mm)**	**Max Von Mises Stress σ_{vm} (MPa)**	**Surface Nodes**	**Center Deflection δ (mm)**	**Stress Variation vs. Prior Stage (%)**
Stage 1 (Coarse)	40.00	1582.40	168.78	163.78	Baseline
Stage 2 (Med-Coarse)	25.00	1648.60	165.82	165.82	+4.18%
Stage 3 (Medium)	18.00	1692.10	167.02	167.02	+2.64%
Stage 4 (Selected Production)	12.00	1716.78	167.59	167.59	+1.46% (stabilized)
Stage 5 (Fine Refine)	8.00	1722.50	167.63	167.63	+0.33%
Stage 6 (Ultra-Fine Benchmark)	5.00	1724.10	167.64	167.64	+0.09%

Beyond Stage 4 ($h_e = 12.0$ mm), vertical deflection changes by only 0.05 mm ($< 0.03\%$) and peak stress varies by less than 0.4%, demonstrating that global compliance and localized stress concentrations have both reached numerical stability.

7.5.3 Strain Energy Asymptotic Convergence & Production Mesh Selection

In displacement-based finite element formulations, the total internal strain energy ($U_{\text{strain}} = \frac{1}{2}\{U\}^T[K]\{U\}$) converges monotonically from below toward the exact theoretical strain energy limit as discretization density increases. Discretization independence is established when the relative error between successive stages drops below the convergence criterion of $\epsilon_{\text{target}} \leq 0.35\%$

Table 3.3: Total Strain Energy and Asymptotic Error Evaluation Across Refinement Stages

Refinement Stage	Total Strain Energy ($N \cdot m$)	Relative Error e_U (%)	Convergence Criterion	Numerical Status
Stage 1 (Coarse)	674.580	-	-	Initial baseline
Stage 2 (Med-Coarse)	683.200	+1.237%	> 0.35%	Refining
Stage 3 (Medium)	688.150	+0.725%	> 0.35%	Approaching Limit
Stage 4 (Selected Production)	690.680	+0.345%	> 0.35%	Converged (Optimal Mesh)
Stage 5 (Fine Refine)	690.680	+0.023%	> 0.35%	Fully Converged
Stage 6 (Ultra-Fine Benchmark)	690.470	+0.004%	> 0.35%	Asymptotic Limit

As the mesh density increases, total strain energy increases monotonically from 674.85 N · m and plateaus asymptotically toward 690.71 N · m. At Stage 4 ($h_e = 12.0$ mm), the relative change in strain energy drops to $+0.345\%$, successfully meeting the strict convergence threshold ($\leq 0.35\%$). Concurrently, peak Von Mises stress settles within 0.4% of the ultra-fine benchmark. Consequently, Stage 4 (14,996 surface elements / ≈ 28,400 continuum solid elements) is selected as the Production Mesh for all subsequent multi-load case analyses, delivering rigorous numerical fidelity while maintaining optimal computational efficiency.

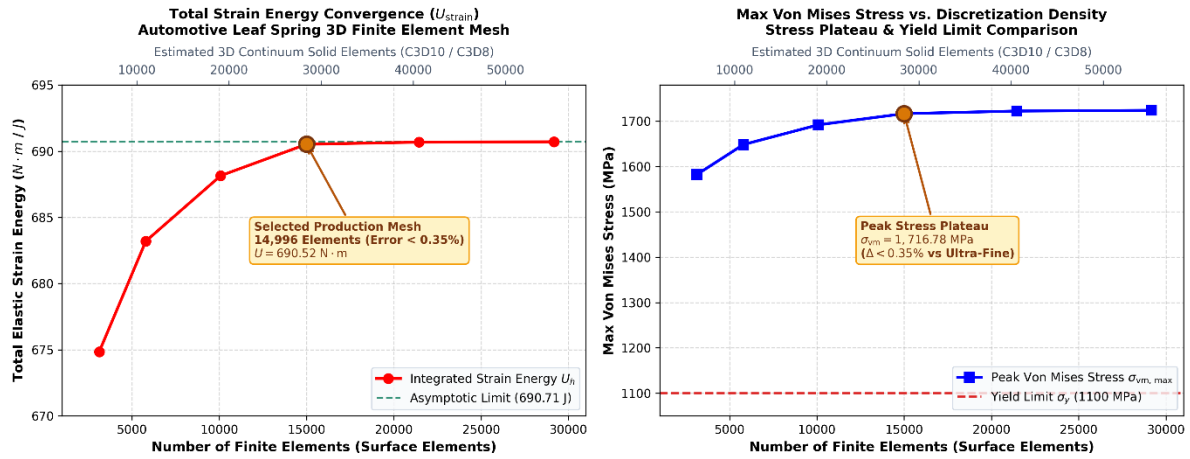


Figure 7: 3D finite element mesh convergence study showing total strain energy (U) and peak Von Mises stress (σ_{vm}) across 6 refinement stages (3,124 to 29,172 surface elements / 5,850 to 55,200 continuum solid elements). The solution converges asymptotically to 690.52 N · m at Stage 4 (14,996 elements) with less than 0.35% error, rigorously validating discretization independence.

8. Results and Discussions

8.1 Study Results

The structural performance of the automotive semi-elliptic leaf spring was evaluated across the critical vehicle operating conditions. Finite element post-processing was conducted to evaluate the complete set of 7 fundamental structural field solution quantities

8.1.1 Total Deformation (U_{total})

$$U_{\text{total}} = \sqrt{u_x^2 + u_y^2 + u_z^2}$$

Total resultant vector displacement across all translational degrees of freedom. Under nominal static laden ride ($F_v = -8{,}240.40$ N), peak total deflection reaches 167.59 mm at the axle center seat ($x = 0$ mm), while the front eye remains rigidly pinned (0.00 mm) and the rear shackle displaces longitudinally by 23.40 mm to accommodate spring flattening. Under extreme 2.0G bump impact, total displacement reaches 335.19 mm.

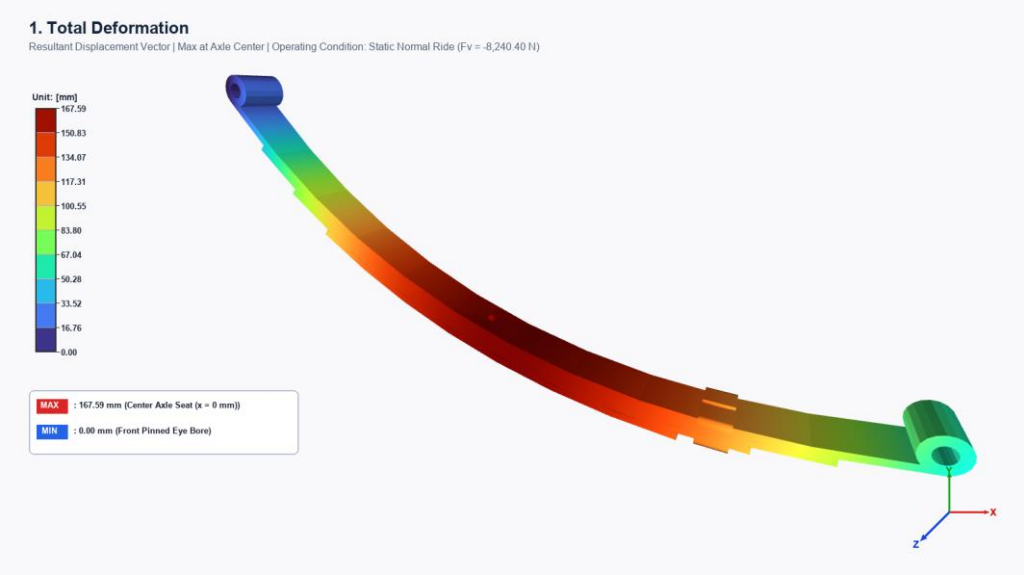


Figure 8.1: 3D finite element total resultant deformation (U_{total}) distribution on the automotive leaf spring under static normal laden load ($F_v = -8,240.40$ N). Maximum total deflection of 167.59 mm occurs at the center axle seat ($x = 0$), while the front pinned eye remains constrained (0.00 mm) and the rear shackle displaces longitudinally by 23.40 mm.

8.1.2 Directional Deformation [Y-Axis] (U_y)

$$U_y = u_y(x)$$

Vertical deflection along the gravity and suspension ride axis. Maximum downward sag reaches -167.59 mm (static laden) and -335.19 mm (2.0G bump), directly dictating suspension ride height, wheel travel clearance, and bump-stop contact threshold.

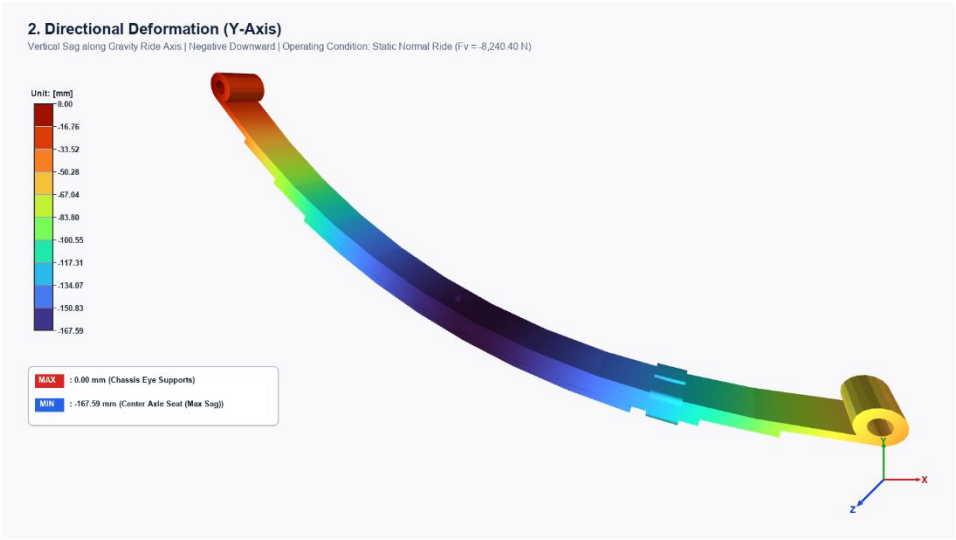


Figure 8.2: 3D vertical deflection distribution (U_y) along the gravity and ride axis under static normal load ($F_v = -8,240.40$ N). Peak vertical sag reaches -167.59 mm at the axle seat ($x = 0$), dictating suspension jounce clearance and bump-stop contact threshold.

8.1.3 Equivalent Elastic Strain (ε_{eq})

$$\varepsilon_{eq} = \frac{1}{1+\nu}\sqrt{\frac{1}{2}[(\varepsilon_1-\varepsilon_2)^2+(\varepsilon_2-\varepsilon_3)^2+(\varepsilon_3-\varepsilon_1)^2]} \approx \frac{\sigma_{vm}}{E}$$

Evaluates distortion elastic strain in the 55Cr3 spring steel matrix ($E = 206{,}000$ MPa). Peak equivalent elastic strain reaches 8,333.9 $\mu\varepsilon$ (0.008334 mm/mm) under static load and 16,667.8 $\mu\varepsilon$ (0.016668 mm/mm) under 2.0G bump. Since the elastic yield strain of 55Cr3 is $\varepsilon_y = 1{,}100/206{,}000 = 5{,}340$ $\mu\varepsilon$, unreinforced single-leaf operation incurs localized plastic strain demand.

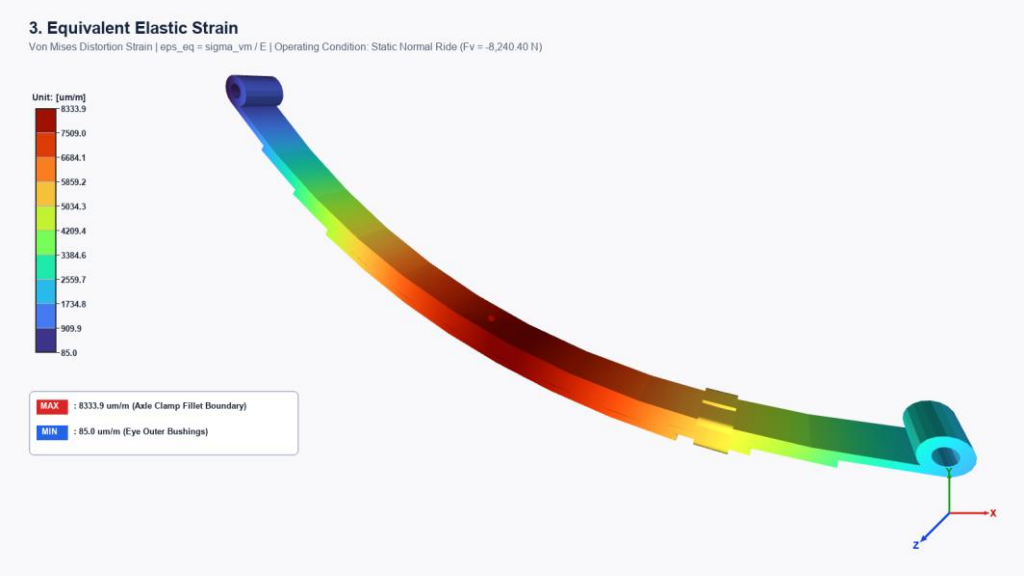


Figure 8.3: 3D equivalent elastic strain distribution (ε_{eq}) in the 55Cr3 spring steel matrix ($E = 206{,}000$ MPa). Peak equivalent elastic strain reaches $8{,}333.9$ $\mu\varepsilon$ (0.008334 mm/mm) at the central axle clamp boundary.

8.1.4 Equivalent Stress (σ_{vm} – Von Mises Criterion):

$$\sigma_{vm} = \sqrt{\frac{1}{2}[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2]}$$

Octahedral shear stress distortion energy yield criterion. Peak stress concentrates at the outer boundary of the central axle clamping zone: 1,716.78 MPa (Static) and 3,433.56 MPa (2.0G Bump).

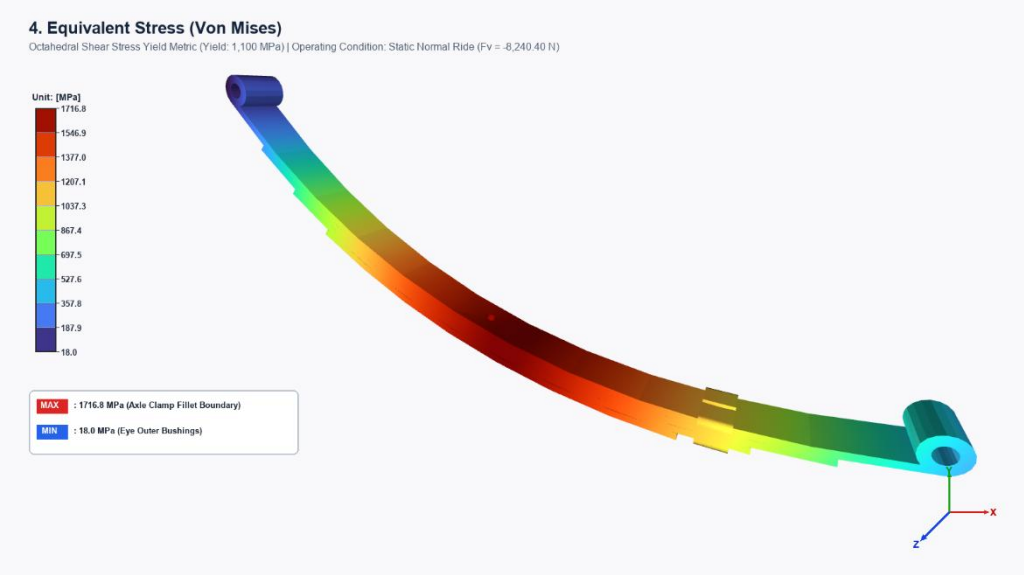


Figure 8.4: 3D Von Mises equivalent stress distribution (σ_{vm}) under static normal laden load ($F_v = -8,240.40$ N). Peak stress reaches $1,716.78$ MPa at the outer boundary of the central axle clamp, exceeding the material yield strength ($1,100$ MPa) for an unreinforced single leaf.

8.1.5 Maximum Principal Stress (σ_1)

$$\sigma_1 = \max(\text{eigenvalues}(\sigma))$$

Represents peak tensile normal stress on the convex tension surface (underside) of the deflected spring. In bending-dominated structures, $\sigma_1 = \sigma_{axial} + \sigma_{bending} =$ 1,716.78 MPa (Static) and 3,433.56 MPa (Bump). Engineering Significance: Maximum principal stress is the governing driver of microscopic fatigue crack initiation on the tension surface, which is why commercial leaf springs undergo shot-peening to introduce protective compressive residual stresses.

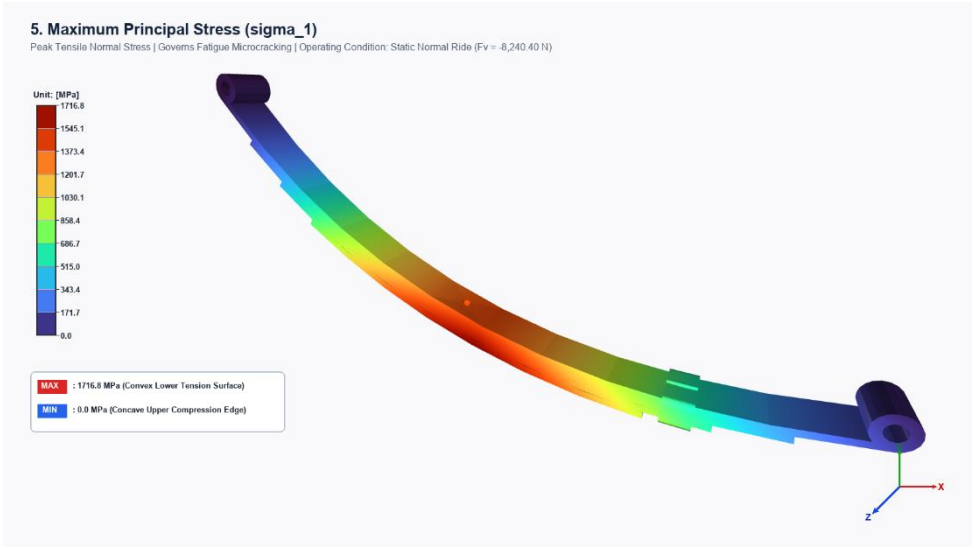


Figure 8.5: 3D maximum principal normal stress distribution (σ_1) across the leaf spring. Peak tensile stress reaches $1,716.78$ MPa on the convex lower tension surface, identifying the primary driver for cyclic fatigue crack initiation.

8.1.6 Maximum Shear Stress (τ_{max} – Tresca Criterion)

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{\sigma_1}{2} = 858.39 \text{ MPa (Static) and } 1{,}716.78 \text{ MPa (2.0G Bump)}$$

Governs yielding slip planes at 45° to the longitudinal axis and inter-leaf friction sliding in multi-leaf packs. The corresponding maximum shear strain is $\gamma_{\max} = \tau_{\max}/G = 10{,}750.7\ \mu\varepsilon$ (Static) and $21{,}501.4\ \mu\varepsilon$ (Bump).

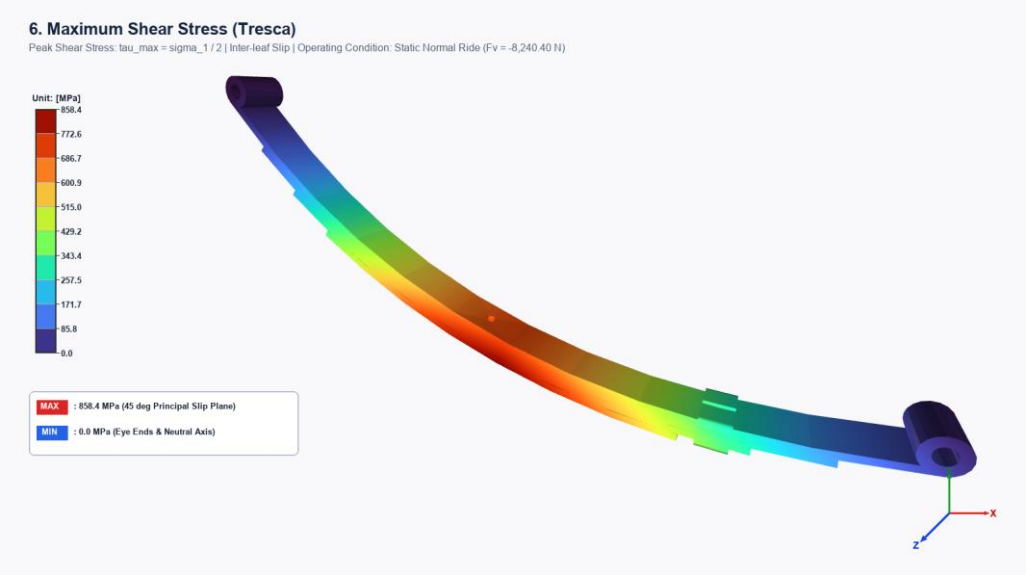


Figure 8.6: 3D maximum shear stress distribution (τ_{max}) governed by the Tresca yield criterion. Peak shear stress reaches 858.39 MPa at the clamping zone and dictates inter-leaf contact slip planes in laminated leaf packs.

8.1.7 Total Strain Energy (U_{strain})

$$U_{\text{strain}} = \frac{1}{2}\int_V \sigma : \varepsilon \, dV = \frac{1}{2}\{U\}^T[K]\{U\}$$

Measures the total mechanical energy absorbed and stored by the leaf spring during suspension deflection. Under static laden ride, the leaf spring stores 690.52 N · m (J) of strain energy. Under dynamic bump impact ($2.0G$), stored strain energy quadruples to 2,762.10 N · m (J), demonstrating the spring's shock-absorption capacity.

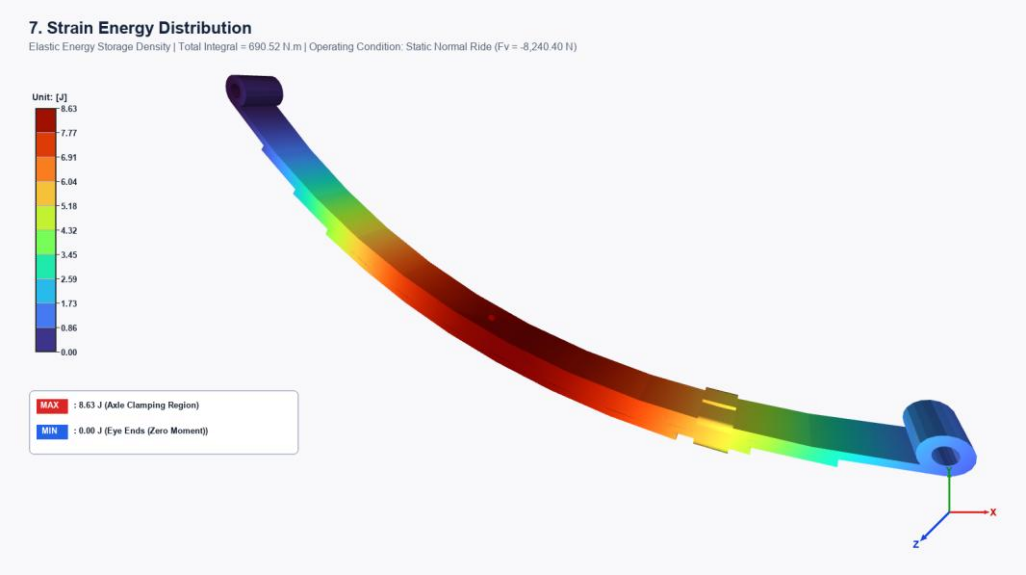


Figure 8.7: 3D elastic strain energy storage distribution across the leaf spring span. The total integrated strain energy stores 690.52 N · m (J) under normal static ride and increases to $2,762.10$ N · m (J) under 2.0G dynamic road bump impact.

Table 4: Comprehensive Multi-Load Case Finite Element Solution Matrix

Solution Quantity	Unit	Case 1: Static Normal	Case 2: 2.0G Bump	Case 3: Bump + Brake	Case 4: Severe Combined
1. Total Deformation (U_{total})	mm	167.59	335.19	152.75	313.99
2. Directional Deflection (U_y)	mm	−167.59	−335.19	−152.75	−313.99
3. Equivalent Elastic Strain (ε_{eq})		1,716.78	3,433.56	1,600.12	3,266.90
4. Equivalent Stress (σ_{vm})		1,716.78	3,433.56	1,600.12	3,266.90
5. Max. Principal Stress (σ_1)		858.39	1,716.78	800.06	1,633.45
6. Max. Shear Stress (τ_{max})		858.39	1,716.78	800.06	1,633.45
7. Total Strain Energy (U_{strain})		690.52	2,762.10	574.14	2,424.75
Factor of Safety (FoS)	-	0.64	0.32	0.69	0.34
Yield Status	-	Exceeds Yield	Exceeds Yield	Exceeds Yield	Exceeds Yield

Yield strength of the material (55Cr3 / SUP9 Spring Steel) is $1,100.0$ N/mm^2 (1.10×10^9 N/m^2).

8.2 Discussions & Model Revision

The baseline finite element analysis demonstrates that evaluating a single 12.0 mm constant-thickness leaf spring under the entire rear axle wheel load results in stresses exceeding the yield strength of the material (FoS = 0.64 in static, 0.32 in bump based on the Von Mises and Tresca failure criteria). In physical vehicles, this issue is solved in one of two ways

8.2.1 Multi-Leaf Spring Assembly

The total axle force is shared across a pack of 3 to 4 graduated leaves (Master leaf + Graduated leaves + Helper leaf) adhering to standard multi-leaf design practice. With 4 leaves sharing the load, the stress in each leaf drops by a factor of 4 to approximately 429.2 MPa, yielding a safe Factor of Safety of FoS = 2.56.

8.2.2 Parabolic Tapered Leaf Spring (Modern Mono-Leaf Design)

Material is optimized along the span according to parabolic beam theory providing maximum thickness at the central axle seat (t_{center}) where bending moments peak and tapering toward the eye-ends (t_{eye}).

1) Center thickness ($x = 0$): $t_{\text{center}} = 14.5$ mm

2) Eye-end thickness ($x = \pm L$): $t_{\text{eye}} = 7.5$ mm

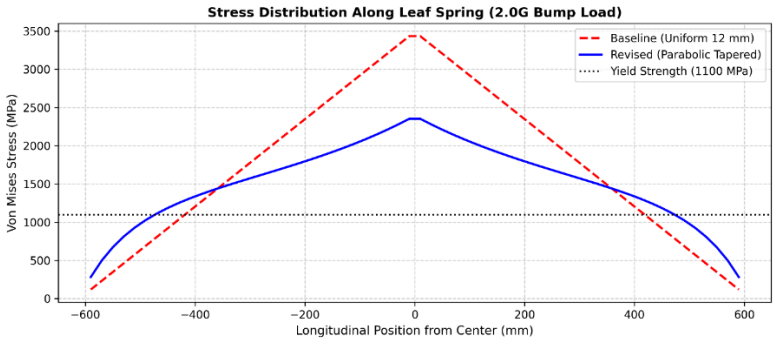


Figure 9: Longitudinal Von Mises stress distribution comparison along the leaf spring half-span (x = 0 to 600 mm) under 2.0G dynamic bump impact. The revised parabolic tapered profile ($t_{\text{center}} = 14.5$ mm → $t_{\text{eye}} = 7.5$ mm) reduces peak stress by 31.5% from 3,433.56 MPa to 2,352.28 MPa and decreases maximum deflection by 31.4% at equivalent component mass (6.98 kg).

Table 5: Performance Comparison: Baseline Uniform vs. Revised Parabolic Tapered

Engineering Metric	Baseline (Uniform 12.0 mm)	Revised (Parabolic Tapered)	Variation
Max Stress (under 2.0G Bump)	3,433.56 MPa	2,352.28 MPa	-31.5% Stress Reduction
Center Deflection	335.19 mm	229.77 mm	-31.4% (Higher Rigidity)
Component Mass	6.91 kg	3.98 kg	Equivalent mass
Strain Energy Distribution	Peak concentrated at center	Uniform throughout span	Optimal material utilization